

Kenneth Udovic

Senior Software Engineer · Full-Stack & Internal Platforms

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Senior software engineer with 9 years building and operating production web applications, focused on internal platforms and workflow systems that replace manual processes. Lead engineer for a university HR hiring and onboarding platform (Next.js, Postgres, Turborepo) with federated OIDC single sign-on, automated contract generation, and queue-backed background processing. Owns systems end to end — relational data modeling, REST APIs and integrations, authentication and role-based access, audit logging, and CI/CD — and serves as the direct technical contact for non-technical stakeholders in compliance-driven environments including healthcare EMR and higher education.

EXPERIENCE

Senior Applications Developer — Rutgers University, Division of Continuing Studies

Apr 2018 – Present

Lead full-stack engineer · sole technical contact for non-technical leadership · owns code review and mentors junior and student developers.

- Lead engineer for an internal HR hiring and onboarding platform built as a Turborepo monorepo (Next.js, Postgres) — dynamic shared forms, automated contract generation, DocuSign integration, CLI tools, cron, and queue-backed background jobs — replacing a manual, paper-based process.
- Deployed and manage a Keycloak instance federated with university identity providers and user federations, giving applications OIDC single sign-on and role-based access against central authentication via NextAuth.
- Architected a Dockerized ETL pipeline that extracts, transforms, and compresses an Oracle database into a binary datafile, enabling performant offline use of a PWA front end.
- Built Elasticsearch-backed paginated search with a Laravel-driven tag filtering system, and front ends rendering multimillion-row datasets.
- Integrated PDF invoice generation and third-party payment processing, and coordinated Node.js WebSocket servers with Redis and PHP for a real-time event system.
- Ran build-versus-buy evaluations and replaced paid vendors with customized open-source deployments, saving the department \$70,000+ per year.
- Introduced containerized deployments and CI/CD using Podman, systemd, GitHub Actions, and Rundeck; stood up multi-server AWS (EC2) hosting for WebRTC classrooms serving 400+ attendees.
- Apply OpenSSL and GPG for flat-file encryption; maintain build tooling and linting standards across PHP and TypeScript.

Software Engineer — Fusion Health

Aug 2016 – May 2017

Healthcare software for correctional facilities — a regulated, audit-sensitive environment.

- Built full-stack single-page applications integrating GE Healthcare's Centricity EMR for medical facilities in correctional environments.
- Created a templating, publishing, and encryption system for legal EMR documents, with digital and fingerprint signature capture.
- Scripted data-migration software against EMR pharmaceutical datasets, ensuring accurate transfer of inmate prescription records.

Technical Intern — Processor Innovations Corporation

Jan 2015 – Aug 2016

- Built a mapping application for calculating and graphing antenna signal strength; performed business-intelligence data analysis; configured Linux servers.

SELECTED PROJECTS

AlmanacRAG — Author

TypeScript · PostgreSQL · Terraform

github.com/Handfish/AlmanacRAG — live: almanac-rag.pages.dev

- Designed and deployed a fact-grounded retrieval system over a live 995-page course catalog: the model selects which rows to show and Postgres supplies every price, date, and status, so incorrect facts are structurally impossible rather than tuned away.
- Built hybrid retrieval — vector kNN and BM25 fused by Reciprocal Rank Fusion in one SQL round trip — behind a typed natural-language-to-filter router, so structured predicates resolve deterministically.
- Provisioned infrastructure as code in Terraform (Neon, Google Cloud Run, Cloudflare Pages) with CI-driven deploys from GitHub Actions; 160+ tests including Testcontainers against a real Postgres per suite.
- Documented the architecture and the decisions behind what was deliberately not built, each grounded in measurements against the live corpus.

talos-pilot — Author & Maintainer

Rust · gRPC · Kubernetes

github.com/Handfish/talos-pilot — 150+ stars · shared by Sidero Labs (makers of Talos Linux)

- Wrote a Talos gRPC machine-API client from scratch in Rust and a terminal UI for operating bare-metal Kubernetes clusters — node monitoring, log streaming, etcd quorum health, and workload diagnostics.
- Fixed multi-node control-plane routing and merged an outside contributor's fix for request-forwarding failures — real bug reports from operators running the tool on their own clusters.
- Wrote all cluster operations to an audit log; shipped via Homebrew tap, shell and PowerShell installers, and a Nix flake.

SKILLS

Languages TypeScript, Rust, PHP, C#, SQL, Bash

Backend & Data Node.js (Effect, Hono, Express), Next.js, Laravel, PostgreSQL / pgvector, MySQL, SQLite, MS SQL Server, Oracle, Elasticsearch, Redis, REST APIs, WebSockets, webhooks, ETL

Security & Identity Keycloak / OIDC, NextAuth, role-based access control, audit logging, OpenSSL / GPG encryption, digital signatures

Frontend React / React Native, Next.js, Svelte, Vue, Astro, Tailwind CSS, Radix UI, Storybook, Cypress

Infrastructure & CI/CD Docker / Podman, systemd, GitHub Actions, Rundeck, Terraform, Pulumi, AWS (EC2/ECS/EKS/Lambda), Google Cloud Run, Kubernetes, Nginx / Caddy / Apache

Practices Monorepos (Turborepo), microservices, CI/CD, infrastructure as code, testing (vitest, Testcontainers, Cypress), technical design documentation, mentoring

EDUCATION

B.S. Computer Science, Minor in Entrepreneurship May 2016

Rutgers University – New Brunswick · GPA 3.4